

Bumblebees like flowers...presumably?

Jenna B. Melanson^{1*}, Natalia Clermont², Tyler T. Kelly², Claire Kremen^{1,2}

1. Biodiversity Research Center, University of British Columbia, 2212 Main Mall, Vancouver, BC, Canada, V6T 1Z4
2. Institute for Resources, Environment, and Sustainability, University of British Columbia, 2202 Main Mall, Vancouver, BC, Canada, V6T 1Z4

★ Corresponding author: Jenna B. Melanson (jenna.melanson@ubc.ca)

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Data Availability: Version-controlled code necessary to reproduce the analyses can be found at <https://github.com/jmelanson98/fvpollens>. Full datasets will be shared to the same repository and archived on Dryad following acceptance.

1

Abstract

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abstract text

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Keywords: keyword1, keyword2, keyword3

1 Introduction

intro text here

2 Methods

2.1 Study System

2.2 *Bombus* and Floral Surveys

2.3 Pollen Metabarcoding

2.3.1 Pollen Collection and Quantification

For all *B. mixtus* and *B. impatiens* workers, we recorded the presence or absence of corbicular pollen as a binary variable. We then removed pollen from the corbiculae of all *B. mixtus* workers and a subset of *B. impatiens* workers; *B. impatiens* workers were selected from the time period which maximized overlap with *B. mixtus* worker activity (18 July-16 August) (Fig. A1). This selection of samples allowed us to draw conclusions about season-long and interannual variation in *B. mixtus* pollen collection, while also allowing us to make comparisons between the two species—which have differing phenologies—during a period of temporal overlap when resource availability was the same for both species.

We measured the mass of collected pollen loads using a microbalance (GET BRAND) and stored the pollen at -80°C until DNA extraction. In cases where only a single pollen load was present (e.g., because the second pollen load was dislodge during bee collection) we multiplied the mass of the single pollen load by two to estimate the total mass of pollen collected by the bee.

25 2.3.2 DNA Extraction, Amplification, and Sequencing

26 2.3.3 Bioinformatics

27 We processed ITS2 amplicon sequencing data using the DADA2 pipeline (Callahan,
28 n.d.; Callahan et al., 2016) implemented in R version 4.3.3 (R Core Team, 2021) with li-
29 braries *dada2* (Callahan et al., 2016), *Biostrings* (Pagès et al., 2026), and *ShortRead* (Mor-
30 gan et al., 2009). Given the variable length of ITS sequences across species, we filtered
31 the demultiplexed reads using $\text{maxN} = 0$ but no sequence truncation. We used *cu-*
32 *tadapt* (Martin, 2011) to trim primer sequences, then performed a second round of
33 quality filtering using the parameters $\text{minLen} = 50$, $\text{maxN} = 0$, $\text{maxEE} = c(2, 2)$, $\text{truncQ} =$
34 2 (DADA2 default parameters (Callahan, n.d.)). The DADA2 pipeline estimates the se-
35 quence error rates to probabilistically discriminate sequences real sequences from those
36 arising through sequencing errors; we used the *learnErrors* function to generate a
37 parametric model of sequence errors, which was used to perform inference of the
38 assigned sequence variants (ASVs) using the *dada* function (Callahan et al., 2016).
39 Finally, we removed chimeric sequences using the *removeBimeraDenovo* function
40 (Callahan et al., 2016); for all three sequencing runs, the percentage of reads identi-
41 fied as chimeric was between 9-11%

42 The output of the DADA2 pipeline is a table of ASVs (i.e., unique sequences) and their
43 abundances across samples. We assigned taxonomy to ASVs in QIIME2 (Bolyen et al.,
44 2019)

45 2.4 Statistical Analyses

46 Some ideas for analysis:

- 47 • colony-level bumblebee-floral networks (see (Ye et al., 2024)) – but find some
48 way to account for phenology

49 3 Results

50 4 Discussion

51 References

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81 **Supporting Information**